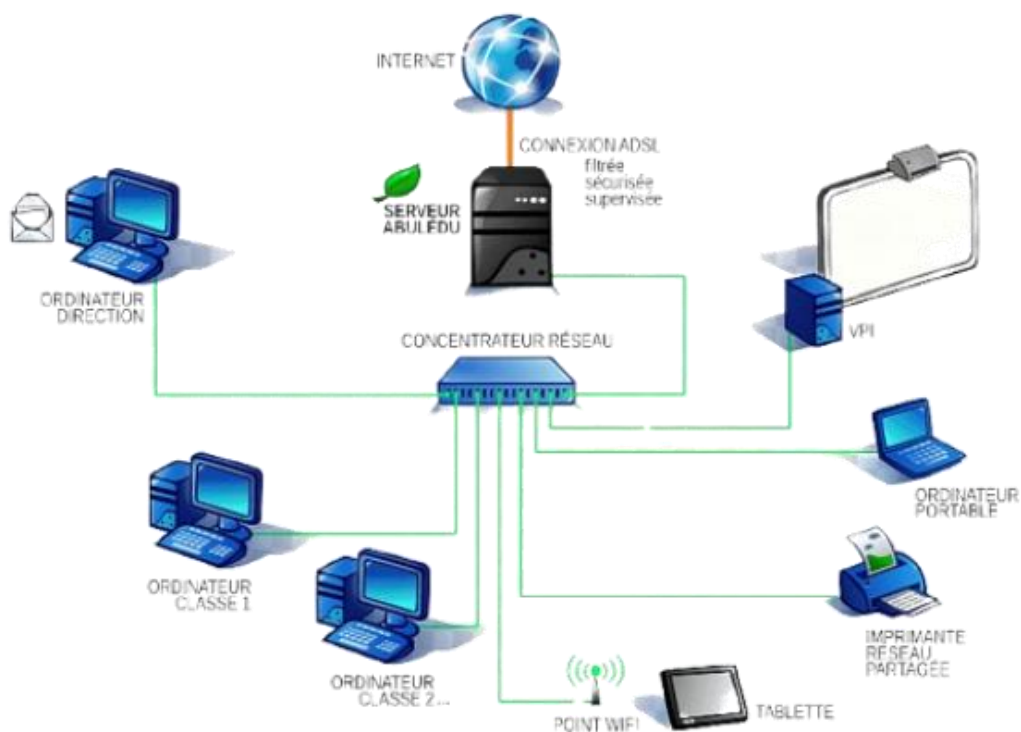


SAE 3.3 — Projet pépinière

Multi-Sites

Rapport technique — Infrastructure réseau (Siège / Succursale)



Auteurs : Adam Tmimi, Anton Maurette

Formation : BUT Réseaux & Télécommunications — IUT de Béthune

Année : 2025–2026

Date : 07/02/2026

1. Synthèse

Dans le cadre de la SAE 3.3, nous avons mis en place une maquette d'infrastructure réseau multi-sites comprenant un siège et une succursale. L'objectif est d'assurer une segmentation par VLAN (ADMIN, PERSONNEL, VIDEO, PRODUCTION, WIFI invité), une administration sécurisée (SSH limité au VLAN ADMIN), une redondance au siège (HSRP), une interconnexion inter-sites via tunnel GRE, un accès Internet contrôlé via proxy au siège, ainsi que les services DHCP, DNS et centralisation des logs (Syslog).

Une procédure de démonstration et les preuves (captures/commandes) sont indiquées pour chaque exigence du cahier des charges.

2. Répartition des tâches

Étudiant	Réalisations principales
Adam Tmimi	VLAN/Trunks, SVI & inter-VLAN, ACL (inter-VLAN + management), SSH, DHCP relay, Syslog, Proxy (tests et preuves), validation générale
Maurette Anton	Tunnel GRE, routage (OSPF/Routes), WiFi (personnel/guest), redondance (HSRP + STP), tests inter-sites, documentation et annexes

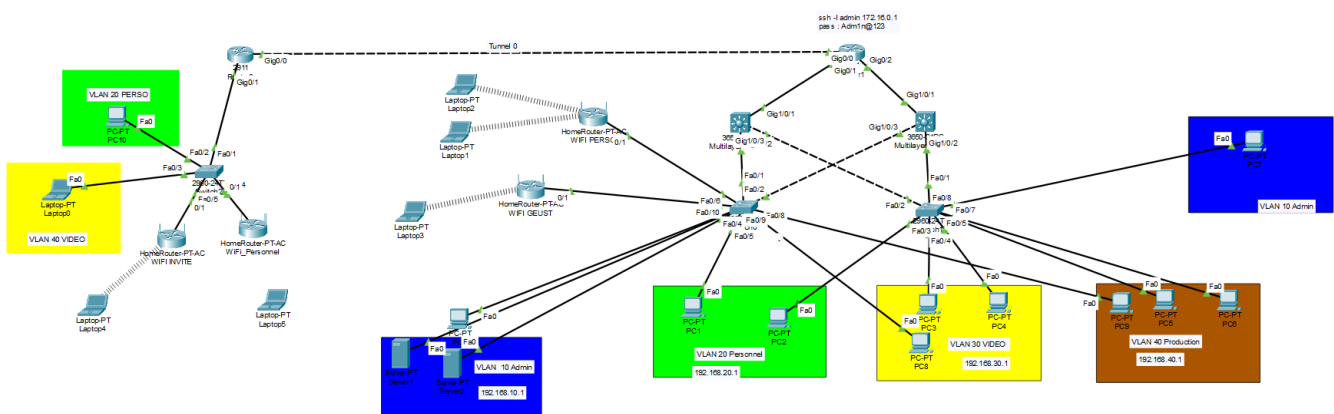
3. Architecture et topologie

3.1 Vue d'ensemble

L'infrastructure est organisée en deux sites :

- **Siège** : cœur réseau (switch L3), segmentation VLAN, serveurs (DHCP/DNS/Syslog), proxy HTTP central, redondance (HSRP).
- **Succursale** : accès utilisateurs segmentés, routage local (router-on-a-stick / sous-interfaces), interconnexion au siège via tunnel GRE.

3.2 Capture de la topologi



4. Plan d'adressage

4.1. Siège – VLAN et passerelles (HSRP .1)

VLAN	Nom	Usage	Réseau	Passerelle
10	ADMIN	Administration/Management	192.168.10.0/24	192.168.10.1 (VIP ou GW)
20	PERSONNEL	Employés	192.168.20.0/24	192.168.20.1
30	VIDEO	Video surveillance	192.168.30.0/24	192.168.30.1
40	PRODUCTION	Zone production	192.168.40.0/24	192.168.40.1
50	GUEST	WiFi invité (Web only)	192.168.50.0/24	192.168.50.1

4.2. Serveurs (Siège – VLAN10)

- SRV-DNS-DHCP : **192.168.10.10**
- SRV-SYSLOG : **192.168.10.11**
- SRV-PROXY : **192.168.10.12**
- PC-ADMIN : **192.168.10.100** (statique)

4.3. Liaisons siège

- R-HQ / CORE-SW1 (L3 /30) : **172.16.0.0/30** (R-HQ .1 / CORE-SW1 .2)
- R-HQ / CORE-SW2 (L3 /30) : **172.16.0.4/30** (R-HQ .5 / CORE-SW2 .6)

4.4. Interconnexion WAN + Tunnel

- WAN R-HQ / R-BRANCH : **192.168.3.0/30** (HQ .1 / BR .2)
- Tunnel GRE : **10.0.0.0/30** (BR .1 / HQ .2)

4.5. Succursale – VLAN (réseaux distincts pour éviter conflit d'adressage)

(Conserve les mêmes VLAN logiques mais avec sous-réseaux différents)

- VLAN20 PERSONNEL : **192.168.120.0/24**, GW **192.168.120.254**
- VLAN30 VIDEO : **192.168.130.0/24**, GW **192.168.130.254**
- VLAN40 PRODUCTION : **192.168.140.0/24**, GW **192.168.140.254**
- VLAN50 GUEST : **192.168.150.0/24**, GW **192.168.150.254**

5. Exigences du cahier des charges

5.1 Réseau ADMIN

Objectif

Créer un VLAN dédié à l'administration permettant l'accès aux équipements et services.

Solution mise en œuvre

- VLAN 10 "ADMIN" configuré sur le siège et la succursale.
- Passerelle L3 (SVI ou sous-interface) pour atteindre les équipements réseau.

Démonstration

- *show vlan brief*

```
ACC-SW-HQ#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/3, Fa0/4, Fa0/10
20	VLAN0020	active	Fa0/5, Fa0/6
30	VLAN0030	active	Fa0/8
40	VLAN0040	active	Fa0/9
50	VLAN0050	active	Fa0/7

- *show ip interface brief*

```
CORE-SW1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1/0/1	172.16.0.2	YES	NVRAM	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	up	up
GigabitEthernet1/0/4	unassigned	YES	unset	down	down
GigabitEthernet1/0/5	unassigned	YES	unset	down	down
GigabitEthernet1/0/6	unassigned	YES	unset	down	down
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	down	down
GigabitEthernet1/0/11	unassigned	YES	unset	down	down
GigabitEthernet1/0/12	unassigned	YES	unset	down	down
GigabitEthernet1/0/13	unassigned	YES	unset	down	down
GigabitEthernet1/0/14	unassigned	YES	unset	down	down
GigabitEthernet1/0/15	unassigned	YES	unset	down	down
GigabitEthernet1/0/16	unassigned	YES	unset	down	down
GigabitEthernet1/0/17	unassigned	YES	unset	down	down
GigabitEthernet1/0/18	unassigned	YES	unset	down	down
GigabitEthernet1/0/19	unassigned	YES	unset	down	down
GigabitEthernet1/0/20	unassigned	YES	unset	down	down
GigabitEthernet1/0/21	unassigned	YES	unset	down	down
GigabitEthernet1/0/22	unassigned	YES	unset	down	down
GigabitEthernet1/0/23	unassigned	YES	unset	down	down
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down
GigabitEthernet1/1/4	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.2	YES	manual	up	up
Vlan20	192.168.20.2	YES	manual	up	up
Vlan30	192.168.30.2	YES	manual	up	up
Vlan40	192.168.40.2	YES	manual	up	up
Vlan50	192.168.50.2	YES	manual	up	up

- *ipconfig* sur le PC ADMIN

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20A:41FF:FE47:1806
    IPv6 Address . . . . . : ::
    IPv4 Address . . . . . : 192.168.10.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : ::
                                192.168.10.1
```

5.2 Administration SSH uniquement depuis VLAN ADMIN

Objectif

Autoriser SSH vers tous les équipements **uniquement** depuis le réseau ADMIN.

Solution mise en œuvre

- Activation SSH sur routeurs/switchs.
- ACL appliquée sur les lignes VTY pour restreindre l'accès au VLAN 10.
Référence démonstration : SSH OK depuis VLAN 10 et refus depuis autre VLAN.

Démonstration

- *show run | section line vty*

```
CORE-SW1# show run | section line vty
line vty 0 4
  access-class 10 in
  login local
  transport input ssh
```

- Capture terminale : SSH OK depuis VLAN 10

```

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::20A:41FF:FE47:1806
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.10.102
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0

C:\>ssh -l admin 172.16.0.2

Password: |

```

- Capture terminale : SSH refusé depuis VLAN 20 (ou autre VLAN)

```

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::201:96FF:FE69:44BE
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.20.103
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                192.168.20.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0

C:\>ssh -l admin 172.16.0.2

% Connection refused by remote host

```

5.3 Réseau PERSONNEL

Objectif

Permettre aux employés (VLAN 20) de communiquer entre eux.

Solution mise en œuvre

- VLAN 20 “PERSONNEL” déployé sur les switches d’accès.
- Les ports utilisateurs affectés au VLAN 20.

Démonstration

- `show vlan brief` (ports affectés au VLAN 20)

```
ACC-SW-HQ#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/3, Fa0/4, Fa0/10
20	VLAN0020	active	Fa0/5, Fa0/6
30	VLAN0030	active	Fa0/8
40	VLAN0040	active	Fa0/9
50	VLAN0050	active	Fa0/7
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

5.4 Réseau PRODUCTION

Objectif

Isoler le trafic PRODUCTION des employés (PERSONNEL) tout en gardant l’accès nécessaire aux services.

Solution mise en œuvre

- VLAN 40 “PRODUCTION”.
- ACL inter-VLAN empêchant PERSONNEL ↔ PRODUCTION.

Démonstration

- `show vlan brief` (VLAN 40 présent)

```
ACC-SW-HQ#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	Fa0/3, Fa0/4, Fa0/10
20	VLAN0020	active	Fa0/5, Fa0/6
30	VLAN0030	active	Fa0/8
40	VLAN0040	active	Fa0/9
50	VLAN0050	active	Fa0/7
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

- Capture ping bloqué : depuis VLAN 20 vers 192.168.40.1 (échec)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

5.5 Réseaux WiFi

Objectif

Deux WiFi (siège + succursale) :

- WiFi personnel (accès interne)
- WiFi invité (Web uniquement)

Solution mise en œuvre

- SSID personnel rattaché VLAN personnel/WIFI interne
- SSID invité rattaché au VLAN GUEST (VLAN 50)
- ACL limitant VLAN GUEST au Web (via proxy)

Démonstration

- Tests :
 - Ping vers VLAN internes depuis invité = échec


```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.1

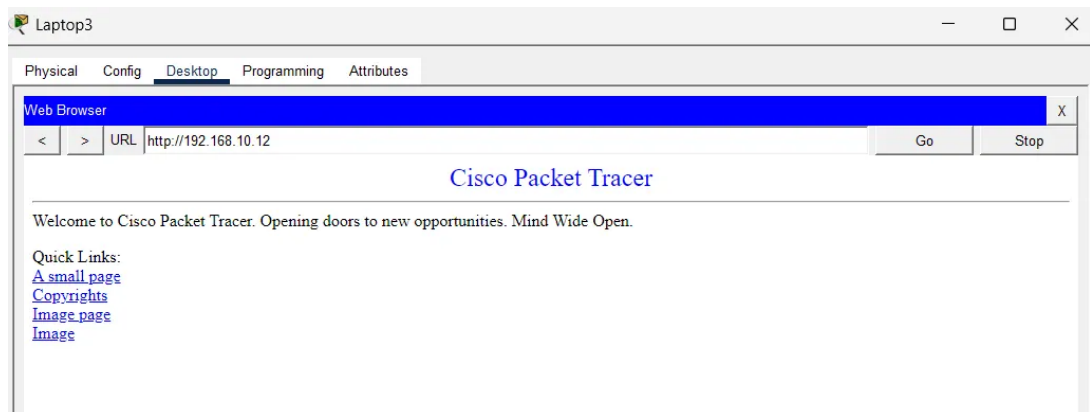
Pinging 192.168.40.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

- Accès au proxy = OK



5.6 Réseau VIDEO privilégié (QoS)

Objectif

Le trafic VIDEO (VLAN 30) doit être prioritaire sur le réseau.

Solution mise en œuvre

- VLAN 30 "VIDEO" configuré.
- Politique QoS appliquée (class-map / policy-map) priorisant VLAN 30.

Démonstration

- show class-map

```

R-BRANCH# show class-map
Class Map match-any class-default (id 0)
  Match any
Class Map match-any CM-VIDEO (id 1)
  Match none
Class Map match-any CM-VIDEO-BR (id 2)
  Match access-group name ACL-VIDEO-BR
Class Map match-any CM-DHCP-BR (id 3)
  Match access-group name ACL-DHCP-BR
- - - - -

```

- show policy-map

```

R-BRANCH#show policy-map
Policy Map PM-WAN-QOS
  Class CM-VIDEO
    Strict Priority
    Bandwidth 30 (%)
  Class class-default
    Flow based Fair Queueing
    Bandwidth 0 (kbps) Max Threshold 64 (packets)
Policy Map PM-QOS-BRANCH
  Class CM-DHCP-BR
    Bandwidth 5 (%) Max Threshold 64 (packets)
  Class CM-VIDEO-BR
    Strict Priority
    Bandwidth 30 (%)
  Class class-default
    Flow based Fair Queueing
    Bandwidth 0 (kbps) Max Threshold 64 (packets)

```

5.7 Routage dynamique — OSPF

Siège

Sur **CORE-SW1** :

show ip ospf neighbor

```
CORE-SW1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	1	FULL/DR	00:00:31	172.16.0.1	GigabitEthernet1/0/1

```
CORE-SW1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	1	FULL/DR	00:00:37	172.16.0.1	GigabitEthernet1/0/1

```

CORE-SW1#show ip ospf
Routing Process "ospf 1" with ID 1.1.1.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 6
    Area has no authentication
    SPF algorithm executed 5 times
    Area ranges are
    Number of LSA 6. Checksum Sum 0x038035
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

```

- Voisins OSPF établis avec R-HQ et CORE-SW2

Succursale

Sur **R-BRANCH** :

show ip route

```

R-BRANCH# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Tunnel0
L       10.0.0.1/32 is directly connected, Tunnel0
C       10.255.255.1/32 is directly connected, Loopback10
C       172.16.0.0/30 is subnetted, 2 subnets
O       172.16.0.0/30 [110/1001] via 10.0.0.2, 00:32:02, Tunnel0
O       172.16.0.4/30 [110/1001] via 10.0.0.2, 01:24:52, Tunnel0
C       192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/30 is directly connected, GigabitEthernet0/0
L       192.168.3.2/32 is directly connected, GigabitEthernet0/0
C       192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.10.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
C       192.168.10.253/32 is directly connected, Loopback0
O       192.168.20.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
O       192.168.30.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
O       192.168.40.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
O       192.168.50.0/24 [110/1002] via 10.0.0.2, 00:33:19, Tunnel0
C       192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.120.0/24 is directly connected, GigabitEthernet0/1.20
L       192.168.120.254/32 is directly connected, GigabitEthernet0/1.20
C       192.168.140.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.140.0/24 is directly connected, GigabitEthernet0/1.40
L       192.168.140.254/32 is directly connected, GigabitEthernet0/1.40
C       192.168.150.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.150.0/24 is directly connected, GigabitEthernet0/1.50
L       192.168.150.254/32 is directly connected, GigabitEthernet0/1.50
S*    0.0.0.0/0 [1/0] via 10.0.0.2

```

- Routes vers les réseaux du siège apprises via OSPF
- Route par défaut vers le tunnel GRE

5.8 Connexion sécurisée Siège – Succursale (Tunnel GRE)

Objectif

Relier les sites via une interconnexion sécurisée.

Solution mise en œuvre

- Tunnel GRE entre routeur siège et routeur succursale.
- Vérification interface tunnel UP/UP.

Démonstration

- Sur siège : show interface tunnel0

```

R-HQ#show interface tunnel0
Tunnel0 is up, line protocol is up (connected)
Hardware is Tunnel
Internet address is 10.0.0.2/30
MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 192.168.3.1 (GigabitEthernet0/0), destination 192.168.3.2
Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
Tunnel TTL 255
Fast tunneling enabled
Tunnel transport MTU 1476 bytes
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 56 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    726 packets input, 53840 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

```

- Sur succursale : show interface tunnel0

```

R-BRANCH#show interface tunnel0
Tunnel0 is up, line protocol is up (connected)
Hardware is Tunnel
Internet address is 10.0.0.1/30
MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 192.168.3.2 (GigabitEthernet0/0), destination 192.168.3.1
Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
Tunnel TTL 255
Fast tunneling enabled
Tunnel transport MTU 1476 bytes
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)
Last input never, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 57 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
    689 packets input, 55530 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out

```

5.9 Trafic Succursale passant par le siège

Objectif

Tout le trafic externe de la succursale doit remonter au siège et passer par le proxy.

Solution mise en œuvre

- Routage forçant la sortie Internet via siège
- Utilisation du proxy du siège depuis la succursale

Démonstration

- tracer 192.168.10.12 depuis un poste succursale (passage par siège)

```
C:\>tracert 192.168.10.12

Tracing route to 192.168.10.12 over a maximum of 30 hops:

  1    3 ms    0 ms    0 ms    192.168.120.254
  2   11 ms    0 ms   11 ms    10.0.0.2
  3   14 ms   37 ms   14 ms    172.16.0.6
  4    *      *     46 ms    192.168.10.12

Trace complete.
```

5.10 Redondance matériels au siège (HSRP)

Objectif

Tolérance aux pannes au siège.

Solution mise en œuvre

- HSRP sur les VLANs (passerelle virtuelle en .1 ou .254 selon votre choix).
Dans la démo : CORE-SW1 Active / CORE-SW2 Standby.

Démonstration

- *show standby brief* sur les deux équipements

```
CORE-SW1#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State      Active        Standby        Virtual IP
Vl10       10   110  P Active     local         192.168.10.3   192.168.10.1
Vl20       20   110  P Active     local         192.168.20.2   192.168.20.1
Vl30       30   110  P Active     local         192.168.30.2   192.168.30.1
Vl40       40   110  P Active     local         192.168.40.2   192.168.40.1
Vl50       50   110  P Active     local         192.168.50.2   192.168.50.1
CORE-SW1#

CORE-SW2#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State      Active        Standby        Virtual IP
Vl10       10   100  P Standby    192.168.10.2  local          192.168.10.1
Vl20       20   100  P Standby    192.168.20.2  local          192.168.20.1
Vl30       30   100  P Standby    192.168.30.2  local          192.168.30.1
Vl40       40   100  P Standby    192.168.40.2  local          192.168.40.1
Vl50       50   100  P Standby    192.168.50.2  local          192.168.50.1
CORE-SW2#
```

5.10 Proxy HTTP

Objectif

Accès Web via proxy centralisé.

Solution mise en œuvre

- Proxy accessible par DNS et/ou IP
Exemples de démo : <http://proxy.sae.local> ou 192.168.10.12

Preuves :

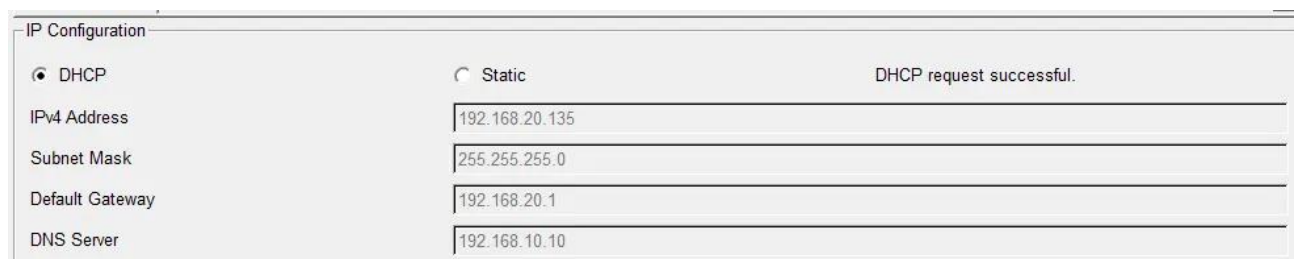


5.11 Service DHCP

Sur un PC du VLAN 20 (PERSONNEL – Siège) :

- Desktop → IP Configuration → DHCP
- IP reçue dynamiquement
- DNS = 192.168.10.10
- Passerelle = 192.168.20.1

(Le DHCP est centralisé sur le serveur du siège)



5.12 Service DNS

Objectif

Résolution des noms internes.

Solution mise en œuvre

- DNS interne (ex : BIND) au siège.
- ping proxy.sae.local

```

C:\>ping proxy.sae.local

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time=28ms TTL=128
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 28ms, Average = 7ms

```

5.13 Centralisation des logs

Objectif

Centraliser les logs des équipements vers un serveur du siège.

Solution mise en œuvre

- Syslog configuré sur équipements et Serveur rsyslog actif au siège

Démonstration

Sur un équipement :

- show logging

```

CORE-SW1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 34 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 34 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 34 message lines logged
  Logging to 192.168.10.11 (udp port 514, audit disabled,
    authentication disabled, encryption disabled, link up),
    34 message lines logged,
    0 message lines rate-limited,
    0 message lines dropped-by-MD,
    xml disabled, sequence number disabled
    filtering disabled

Log Buffer (4096 bytes):

%LINK-3-UPDOWN: Interface Vlan30, changed state to down

```

- show run | include logging

```

R-HQ#show run | include logging
logging 192.168.10.11

```

6. Annexes — Configurations à fournir

Annexe A — VLAN / Trunks

- show run | section vlan

```
CORE-SW1#show run | section vlan
  switchport trunk allowed vlan 10,20,30,40,50
  switchport trunk allowed vlan 10,20,30,40,50
CORE-SW1#
```

- show run interface trunk

```
CORE-SW1#show run | section trunk
  switchport trunk allowed vlan 10,20,30,40,50
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40,50
  switchport mode trunk
CORE-SW1#
```

Annexe B — Routage / Tunnel

- show run interface tunnel0

```
R-HQ#show interface tunnel0
Tunnel0 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 10.0.0.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 192.168.3.1 (GigabitEthernet0/0), destination 192.168.3.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 54 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    400 packets input, 27957 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
```

- show ip route


```
R-HQ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Tunnel0
L       10.0.0.2/32 is directly connected, Tunnel0
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.0.0/30 is directly connected, GigabitEthernet0/1
L       172.16.0.1/32 is directly connected, GigabitEthernet0/1
C       172.16.0.4/30 is directly connected, GigabitEthernet0/2
L       172.16.0.5/32 is directly connected, GigabitEthernet0/2
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/30 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.10.0/24 [110/2] via 172.16.0.2, 00:59:08, GigabitEthernet0/1
        [110/2] via 172.16.0.6, 00:59:08, GigabitEthernet0/2
C       192.168.10.254/32 is directly connected, Loopback0
O       192.168.20.0/24 [110/2] via 172.16.0.2, 00:59:08, GigabitEthernet0/1
        [110/2] via 172.16.0.6, 00:59:08, GigabitEthernet0/2
O       192.168.30.0/24 [110/2] via 172.16.0.2, 00:59:08, GigabitEthernet0/1
        [110/2] via 172.16.0.6, 00:59:08, GigabitEthernet0/2
O       192.168.40.0/24 [110/2] via 172.16.0.2, 00:59:08, GigabitEthernet0/1
        [110/2] via 172.16.0.6, 00:59:08, GigabitEthernet0/2
O       192.168.50.0/24 [110/2] via 172.16.0.2, 00:59:08, GigabitEthernet0/1
        [110/2] via 172.16.0.6, 00:59:08, GigabitEthernet0/2
O       192.168.120.0/24 [110/1001] via 10.0.0.1, 00:59:43, Tunnel0
O       192.168.140.0/24 [110/1001] via 10.0.0.1, 00:59:43, Tunnel0
O       192.168.150.0/24 [110/1001] via 10.0.0.1, 00:59:43, Tunnel0
```

Annexe C — HSRP

- show run | include standby

```
CORE-SW1#show run | include standby
standby 10 ip 192.168.10.1
standby 10 priority 110
standby 10 preempt
standby 20 ip 192.168.20.1
standby 20 priority 110
standby 20 preempt
standby 30 ip 192.168.30.1
standby 30 priority 110
standby 30 preempt
standby 40 ip 192.168.40.1
standby 40 priority 110
standby 40 preempt
standby 50 ip 192.168.50.1
standby 50 priority 110
standby 50 preempt
-----
```

- show standby brief

```
CORE-SW1#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri  P State      Active        Standby        Virtual IP
Vl10           10   110  P Active     local         192.168.10.3   192.168.10.1
Vl20           20   110  P Active     local         192.168.20.2   192.168.20.1
Vl30           30   110  P Active     local         192.168.30.2   192.168.30.1
Vl40           40   110  P Active     local         192.168.40.2   192.168.40.1
Vl50           50   110  P Active     local         192.168.50.2   192.168.50.1
```

Annexe D — ACL / Sécurité

- show access-lists

```
CORE-SW1#show access-lists
Standard IP access list 10
 10 permit 192.168.10.0 0.0.0.255 (8 match(es))
Extended IP access list BLOCK-PROD
 10 deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
 20 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
 30 permit ip any any
Extended IP access list ACL-VLAN30
 10 permit ip 192.168.30.0 0.0.0.255 any
Extended IP access list ACL-DHCP
 10 permit udp any eq bootpc any eq bootps
 20 permit udp any eq bootps any eq bootpc
Extended IP access list GUEST-WEB
 10 permit udp 192.168.50.0 0.0.0.255 host 192.168.10.10 eq domain
 20 permit tcp 192.168.50.0 0.0.0.255 host 192.168.10.10 eq domain
 30 permit tcp 192.168.50.0 0.0.0.255 host 192.168.10.12 eq www
 40 permit tcp 192.168.50.0 0.0.0.255 host 192.168.10.12 eq 443
 50 deny ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
 60 deny ip 192.168.50.0 0.0.0.255 any
```

- show run | section line vty

```
CORE-SW1#show run | section line vty
line vty 0 4
  access-class 10 in
  login local
  transport input ssh
  exec-timeout 0 0
```

Conclusion

La maquette répond au cahier des charges :

- Segmentation VLAN (ADMIN, PERSONNEL, VIDEO, PRODUCTION, GUEST)
- Administration sécurisée (SSH restreint)
- Redondance au siège (HSRP)
- Interconnexion siège–succursale via tunnel GRE
- Trafic succursale sortant via le siège et proxy
- Services DHCP, DNS et centralisation des logs fonctionnels